

Vibhor Vanvani

+91 8377933136 | vibhorvanvani@gmail.com | [LinkedIn](#) | [Github](#) | [Personal Website](#)

WORK EXPERIENCE

D. E. Shaw

Financial Research Analyst, Data Analytics (promoted from Associate in 18 months)

Hyderabad, India

Jun 2024 - Present

Analytics support to traders, portfolio managers, and senior researchers across quantitative research, LLM tooling, and investment decision workflows.

- Corporate Development:** Supported corporate development and strategic decision-making through APAC market intelligence trackers covering liquidity, futures, and activism across 7,000+ instruments; engineered Python dashboards and automated pipelines to deliver LLM-driven market entry analysis.
- LLM Engineering:** Built custom LLM-powered tools - chatbots, OCR pipelines, data parsing and classification systems, and API-integrated extraction across 2,000+ broker reports and earnings calls; reduced manual research effort by ~2 FTE; won D. E. Shaw's international prompt engineering challenge.
- Investment Research:** Optimised holding periods for 6,000+ emerging market bonds across 50 countries via Bayesian regression on Bloomberg and FactSet data, improving accuracy by 70%; Supported research for an activist position on a crypto miner converting to a data center.

EDUCATION

University of Cambridge

Cambridge, United Kingdom

MPhil in Economics and Data Science (Incoming, Oct 2026)

Coursework: Machine Learning in Economics, Causal Inference & ML, Econometric Methods, Microeconomics, Research Computing, Empirical Finance

Recipient of the Cambridge Trust International Scholarship for academic excellence

University of Delhi: Ramjas College

Delhi, India

BA (Hons) Economics | GPA: 8.095/10.0 | Top 10 of 100

Nov 2021 - Jun 2024

Coursework: Econometrics, Data Analysis, Statistics, Linear Algebra, Microeconomics, Macroeconomics

Named to the Dean's Honour List for exceptional academic performance.

Indian Institute of Technology Madras

Chennai, India

Diploma in Data Science and Applications | GPA: 8.02/10.0

Sep 2021 - Dec 2023

Coursework: Machine Learning, Business Analytics, Programming in Python, Statistics for Data Science

Awarded Best Capstone Project in Business Analytics (2023)

INTERNSHIPS

Ernst & Young - Data Analytics Intern

Delhi, India | Jun - Aug 2023

- Applied ML models (Random Forest, Gradient Boosting, SARIMA, Prophet) to trade-flow data across 50+ countries; improved forecast accuracy by 25%; insights shaped national policy on market access and product diversification.

Institute of Economic Growth - Research Intern

Delhi, India | Feb - Mar 2023

- Surveyed 80+ households on anaemia vulnerability; mapped regional hotspots; contributed to policy recommendations on hemoglobinometers in Anganwadis and iron/folic acid in Mid-Day Meals.

Emirates: National Bank of Dubai - Market Analytics Intern

Dubai, UAE | May - Jun 2022

- Selected from India for a Dubai role; analysed uptake of banking and Shariah-compliant products via segmentation and pricing; recommended changes improving product positioning and acquisition strategy.

RESEARCH

V. Vanvani, "[Examining the relation between Calorie Deficiency and Income in Rural India.](#)" The Economics Journal, SRCC, 2022 - NSSO data across 11 income classes; weak income-nutrition correlation (0.21) exposing behavioural gaps in food expenditure.

V. Vanvani, "[Asset Pricing in an Interstellar Economy](#)" (SSRN) - Asset pricing model for delayed-information economies; cointegration tests show persistent price dispersion and a volatility-driven risk premium.

SKILLS

Technical: Python, R, MATLAB, Mathematica, Excel, Tableau, Power BI, Bloomberg Terminal

Certifications: Google Data Analytics Professional Certificate (Coursera, 8-course: spreadsheets, R, Tableau)

PROJECTS

Espresso Protocol - Agentic econometric AI engine: takes a plain-English question and dataset, selects the correct regressor, tests assumptions, auto-corrects on violations, and returns verified results. All computation is deterministic. Terminal CLI with REPL, HTML dashboard, and LaTeX export.

LLMs in a War of Attrition: Built a multi-agent game theory simulation across four U.S. debt-ceiling crises on live real time mako, bonds, and congressional inputs; Quantified bargaining dynamics using regression analysis, showing that relative delay costs strongly predicted concession timing.

Democracy & Carbon Emissions (DU): Multiple regression across nation-states; stronger democratic institutions correlate with lower emissions growth.

POSITIONS OF RESPONSIBILITY

Debating Society: Secretary: Mentored 70+ members to 10+ national wins; founded college's first Debate Equity Committee (+30% participation).

Ramjas Economic Review: Evaluated 30+ economic research papers using data-driven critique, reducing average review cycle from 6 weeks to 2 weeks

Economics Society: Led large-scale economics events with 1,000+ participants, managing end-to-end logistics and budgets

Satcom / ISRO: Assisted the logistics for Indian Space Congress, managing 150+ delegates and supporting research and session coordination

ACHIEVEMENTS

- 1st place at 10+ national debate tournaments across 15 states; chaired finals at 20+ tournaments including international as an invited adjudicator
- Honourable Mention, IMC (International Mathematics Competition) Olympiad for University Students 2022
- 1st Place across three national research conferences of IIT Delhi, IIM Indore, and SRCC.
- Discovered an asteroid through NASA's IASC programme using Pan-STARRS telescope data